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Received: 25 October 2025

Accepted: 14 January 2026

Published online: 21 January 2026

Cite this article as: Bhoopalan R., Mirdula S., Kannusamy P. *et al.* A self attention based deep learning framework for accurate and efficient dental disease detection in OPG radiographs. *Sci Rep* (2026). <https://doi.org/10.1038/s41598-026-36672-2>

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A Self Attention Based Deep Learning Framework for Accurate and Efficient Dental Disease Detection in OPG Radiographs

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Abstract

Oral diseases are increasing now-a-days and there is a high demand for the automatic diagnostic system that helps the clinician to detect these oral diseases with more accuracy and reduced human error. Utilizing the advancement of Deep Learning techniques, this study proposes a novel comparative approach for the diagnosis of teeth diseases using Orthopantomogram (OPG) images and recent transformer based architecture. Particularly, Vision Transformer (ViT) and Swin Transformer are employed for the development of the effective automatic system. Experimental results demonstrated that the Vision Transformer achieved higher performance with a test accuracy of 96%, precision of 95.8%, recall of 96.2%. Swin transformer, with a hierarchical design and shifted window, achieved an accuracy of 95.2% but with efficient inference time

and scalable complexity. Based on the findings, it is inferred that ViT outperforms Swin Transformer in diagnosing oral diseases. Thus the proposed work confirms the effectiveness of transformer based architectures in dental imaging tasks, providing a promisable solution for the clinician, for the automatic diagnosis of oral diseases with high accuracy and less time.

Keywords: Dental Caries, Deep learning, Image Classification, Orthopantomogram Images, Swin Transformer, Vision Transformer.

I. Introduction

Oral diseases lead to a major global health burden, affecting about 3.5 billion people globally [1]. According to the World Health Organization (WHO), it is reported that untreated dental caries in permanent teeth impact about 2 billion individuals, while caries in earlier teeth affect nearly 514 million children worldwide [1]. For the past three decades, the cases of oral diseases have increased by nearly 1 billion, underscoring persistent shortcomings in preventive and therapeutic treatment services [2]. Recent epidemiological evidence also shows that adolescents and young adults in the age group of 10 to 24 are disproportionately affected, accounting more than 643 million cases of oral disorders globally [3]. In addition to that, the incidence of periodical disease and tooth loss, edentulism, has risen sharply in recent days, significantly contributing to Disability Adjusted Life Years (DALYs) and reducing overall quality of life [4].

The oral diseases encompass a wide range of conditions such as Hypodontia, Gingivitis, Caries and Calculus and these are shown in Figure.1.

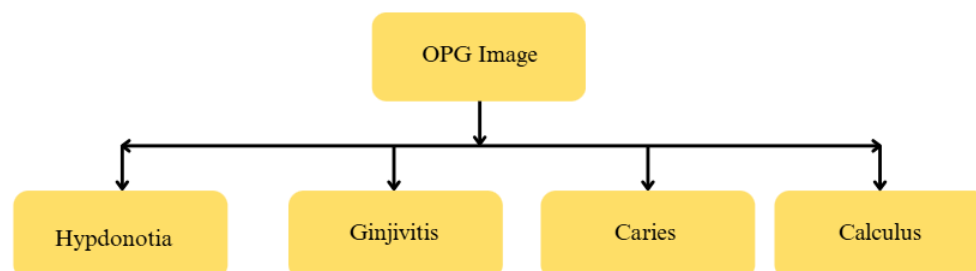


Fig. 1. OPG image and different classes

These diseases are driven by common risk factors including poor oral hygiene, sugar intake, tobacco and alcohol usage, limited fluoride exposure, and socio economic disparities. Among these conditions, dental caries and periodontal disease are most prevalent, frequently resulting in tooth loss if left untreated. Oral cancers, though less prevalent, are associated with high morbidity and mortality, particularly in populations with elevated tobacco and alcohol use [5].

Traditional diagnostic methods primarily rely on visual and tactile clinical examinations, standard indices and radiographic imaging. Clinical inspection and probing are most commonly used methods and easy access methods, but are restricted by subjectivity and examiner variability, particularly in identifying early or hidden lesions [6]. Radiographic techniques including intraoral bitewing and periapical radiographs are frequently used to identify caries and periodontal bone loss, Orthopantomograms (OPGs) offer a panoramic view of maxillofacial area for enabling assessment of tooth positions, gross pathologies and bone structure [7].

Cone Beam Computed Tomography (CBCT) gives 3D imaging and enhanced resolution for periodontal and bone evaluation, through its high cost and greater radiation dose limit routine application [8]. Standard measures like the Decayed, Missing and Filled Teeth (DMFT) index and periodontal probing serve as important tools for epidemiological research and clinical monitoring, though it remains invasive and operator dependent. Adjunctive modalities, including fluorescence imaging and optical coherence tomography, are increasingly researched but have not yet widely implemented in daily clinical practice [9].

Despite traditional diagnostic methods being widely used, they have several limitations. Their effectiveness lacks sensitivity for early stage conditions, and may not provide significant results and may vary across practitioners. Moreover, 2D radiographs can obscure or distort anatomical structure, while modern modalities such as CBCT remain inaccessible in

resource limited areas. These limitations highlight the necessity for automated, scalable and precise diagnostic systems that can support clinical expertise, mitigate human error, and facilitate early intervention.

Orthopantomograms provide an ideal basis for such automation, due to their routine use in dental practices, relatively low radiation dosage, and being able to capture a complete view of teeth and jaw structure in a single scan. Recent technology in Artificial Intelligence, Deep Learning models, mainly transformer based like Vision Transformer (ViT) and Swin Transformer have achieved outstanding performance in medical imaging tasks [10], [11]. This ability to model long range relationships, process scanned images in a hierarchical way, and adapt precisely to varying computational demands makes them well suited for dental imaging. Harnessing these advantages, the current study proposes an automated diagnostic model for teeth disease detection using OPGs, showing that a transformer based framework can deliver high accuracy and efficiency, thereby paving a viable way for clinical practice.

The novelty of the proposed work is that the Vision Based Self Attention Mechanism model achieved a high diagnostic accuracy and will act as a promising clinical tool for the early diagnosis of Oral diseases with further prospective clinical testing.

II. Literature Review

Over the half of the decade, Deep Learning has emerged in dental imaging tasks such as detection, classification, segmentation and numbering. Early studies demonstrated feasibility of Convolutional Neural Networks (CNNs) for detection of tooth and numbering in panoramic radiographs, by establishing automated pipelines that in many routine tasks match the performance of human annotators [12]. In subsequent systematic and narrative reviews, several works have documented steady progress across various tasks like caries detection, tooth segmentation, forensic age estimation, and impacted tooth identification, while also highlighting recurring challenges like annotation variability, dataset heterogeneity,

and lack of multi-centre clinical validation [13], [14]. On the whole, these reviews affirm that Deep Learning can ease the clinician workload and improve reproducibility, and also mention the importance of standard OPG datasets and robust evaluation frameworks to accelerate translation to enable clinical practice.

Work focused on panoramic radiographs (OPGs) has progressed from traditional CNN based methods to more advanced architecture addressing localization, frequent segmentation and various class tooth classification. Recent studies introduced end-to-end models for automated tooth detection, segmentation, and numbering with suitable performance for clinical use, and achieving performance improvements through large annotated datasets and multi class workflows [15], [16]. Other researches targeted disease specific tasks on OPGs like caries or impacted tooth detection, by semantic segmentation and object detection networks to identify lesions and provide quantitative outputs such as lesion area or bone loss measurements. However, much of this research still relies on CNN derivatives and standard augmentation with transfer learning schemes and report challenges in generalization to different centres and acquisition devices.

The Swin Transformer developed extended ViT ideas with a hierarchical framework with shifted window self-attention to achieve linear computational complexity relative to size of the image and effective multi-scale representation that make it highly effective for dense prediction like segmentation and object detection [17]. Surveys of transformers in medical imaging consistently report a gain in classification and segmentation and emphasize improved flexibility for multi scale and multi model challenges, though they also flag increased data requirements and computational tasks as practical demands.

Recent research has started applying transformer based models especially to dental imaging. Comparative research evaluating ViT variants and hierarchical frameworks like Swin on medical images suggest that ViT can achieve state-of-the-art classification with suitable preprocessing and transfer learning, whereas the Swin transformer often excels on dense

prediction problems and multi scale features [18],[19]. In the dental applications, hybrid transfer learning methods integrating CNN encoders with transformer heads or directly fine tuning of ViT / Swin on datasets of OPG, have been proposed. These methods have proposed competitive accuracy for tooth level classification and identification of diseases, often with improved greater resilience to global image variations compared to CNN baselines [20], [21]. Several works also proposed application specific adaptations, with the example of tooth type embedding, multi task heads with disease classification and Numbering, and windowing methods adapted to panoramic image geometry, all together to exploit the transformer's global context modelling while maintaining computational cost.

Although promising results, the literature studies highlight gaps and open problems for clinical translation. Initially, there is a limitation of large, open, and well annotated panoramic datasets that clearly represent diverse demographics, devices, and pathologies. Current dataset heterogeneity limits external validity and hampers comparisons across studies. Second, transformer based approaches inherit the explainability and uncertainty in estimation challenges of Deep Learning frameworks, an important limitation for clinical trust, and some dental research provide limited model interpretation analysis or prospective validation in the real world. Third, computational complexity and inference speed must be balanced with clinical workflow requirements. While Swin Transform offers relatively efficient for high resolution images, optimization is still required for real time or edge deployment.

Finally, more studies rely on retrospective evaluation, multi centre trails and prospective are rare but essential to perform real world safety and effectiveness. These limitations motivate the need for hybrid methods focus on establishment of standard evaluation benchmarks and the present proposed method follows by comparing ViT and Swin transforms models on OPG based dental diagnosis and providing comprehensive metrics like Precision, Recall, Accuracy, F1 score together with inference characteristics.

There has been rapid development in automatic analysis of panoramic dental radiographs, enabled by deep learning techniques. Fang et al. [22], introduced a multi-channel convolution neural network with clinical prior for wisdom tooth detection, and showed that the addition of clinical metadata to image features leads to better diagnostic performance in OPG. While their model showed good performances in the mTm number localization, its CNN structure was still constrained to capture global contextual dependencies from full panoramic images for multi-disease detection. Guiding to clinical states rather than anatomy alone, Iacob et al. [23] reviewed current AI techniques for periodontal bone loss detection and emphasized on challenges like poor generalization in different datasets, intra-observer variability during annotation marks in images, and limitations of conventional convolution methods to fully characterise interpixel relationships among tooth structures and tissues.

With the object detection based models and the rise of the oriented approaches, Şevik and Mutlu [24] utilized a YOLO based single stage model for anomaly detection in pediatric OPGs and obtained real time detection along with promising accuracy. Their findings showed that YOLO can be adapted to localization based tasks, but single stage detectors are based on localized convolutional filters, and cannot effectively handle disease classification applications demanding long range contextual reasoning. In order to break the representational bottlenecks of CNN centric models, Singh et al. [25] embedded transformer attention into a YOLO backbone optimized with genetic algorithms, achieving remarkable progress in anomaly detection and tooth type classification. Their work demonstrates that attention mechanisms contribute to expressive features on dental radiographs, while the hybrid structure preserves detection oriented characteristics instead of learning global image context.

Apart from individual works, a systematic review on automated dental condition detection on panoramic radiographs [26] exemplified the impediments, limitations and future prospects in pursuit of clinically reliable CAD systems. The review emphasized the need for advanced architectures that can perform multi-semantic reasoning across the image

through a global context predictions, demonstrating robustness against variation in imaging condition and supporting simultaneous multi-condition prediction rather than isolated classification tasks. Golkarieh et al. [27] investigated various deep learning for dental condition classification using panoramic X-ray images by comparing a number of CNN and hybrid structures, showing that stronger global modelling feature extractors are required to improve the accuracy and stability across datasets. These studies exhibit a common research gap that there is still absence of models which robustly and efficiently detect multiple oral diseases and model long-range dependencies over OPGs. This shortfall is one of the main driving forces for applying transformer based models like Vision Transformer and Swin Transformer for accurate and efficient diagnostic modelling from PDs.

In summary, the existing documents show a clear progression from CNN driven methods for OPG analysis toward transformer based approaches, which offers modified global context representation and competitive accuracy. The current research offers the comparative evaluation of ViT and Swin transformer pipelines on OPG scans for tooth diagnosis and examines trade-offs between classification accuracy and computational efficiency in clinical relevant contexts.

The Key Contributions of this proposed work are:

- This study presents a novel comparative model employing Vision and Swin Transformer for the early detection of dental diseases by evaluating both global and hierarchical self-attention mechanisms for diagnosis.
- This model provides a complete end-to-end design initializing from pre-processing to transformer based classification with an effective post processing for clinical detection. Thus this design enhances diagnostic autonomous detection as well as preserves medical relevance.
- This research explains a quantitative comparison of performance evaluation by calculating multiple metrics like accuracy, precision, recall, F1-score, ROC and AUC. It is

revealed that ViT achieves 96% of accuracy and good diagnostic reliability.

- The study utilizes a vast dataset of OPG images comprising four classes like Caries, Calculus, Gingivitis and Hypodontia under different imaging conditions. This setup contributes a benchmark approach for future research.
- The proposed research highlights the trade-off between the model accuracy and computational efficiency which offers a practical insight for dental diagnosis in the real world.

III. Proposed Methodology:

Figure 2 shows the proposed flow diagram which illustrates the systematic pipeline for the automatic diagnosis of oral diseases using self-attention mechanisms. The input OPG image is captured, pre-processed and classified by the Vision transformer (ViT) and Swin Transformer. To compare the model efficiency and accuracy, Performance evaluations are calculated and it is discussed in the later part of the paper.

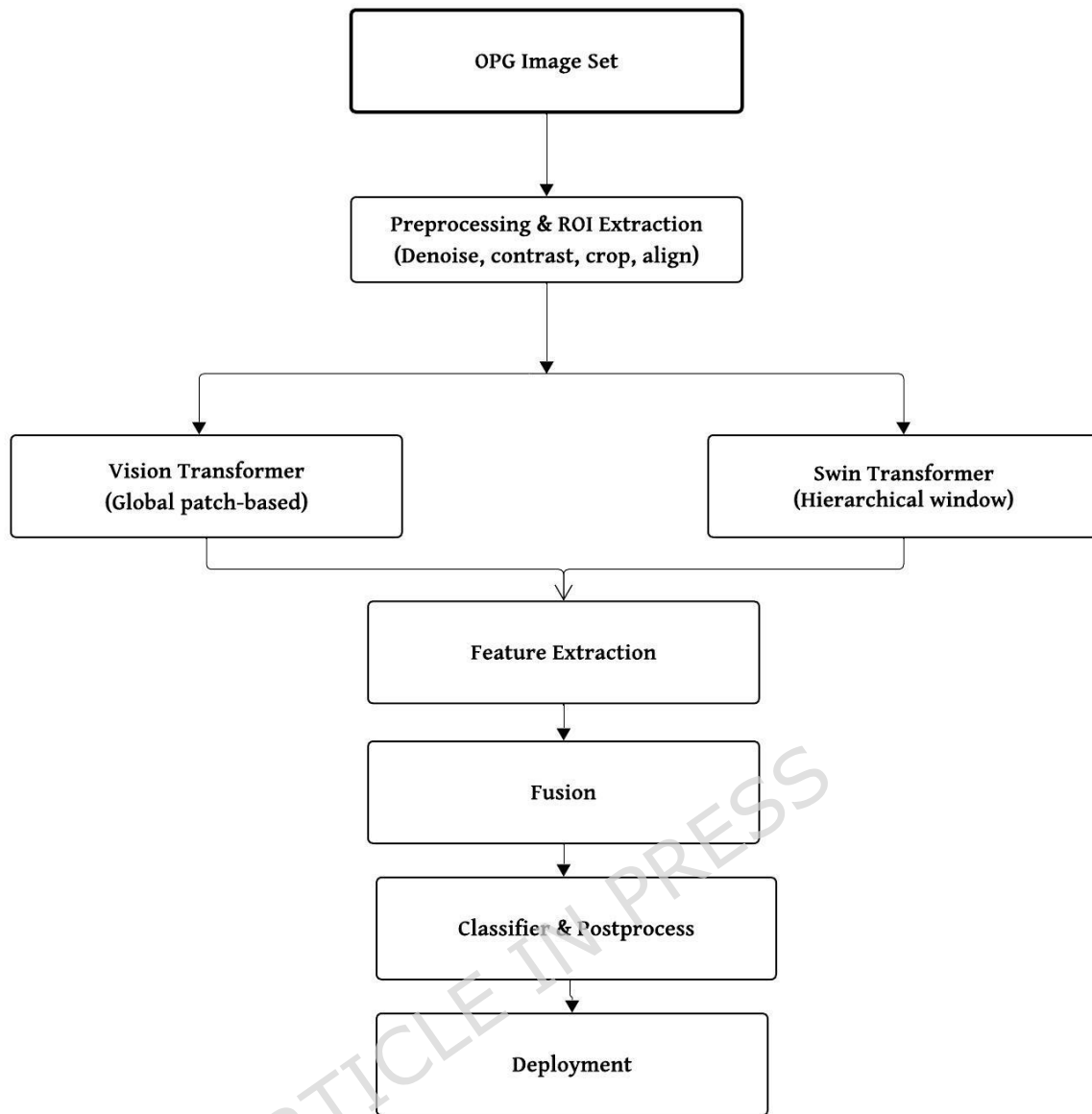


Fig 2. Flow Diagram of the Proposed Method

i) OPG Image Set

In this work, it is proposed to use a public dataset of OPG images for training and testing our method, which includes panoramic dental radiography images from clinical sources and publicly available repositories [28]. The Dataset used in this research article is collected from the Oral Dataset set from Kaggle by Sajid 2024 [29]. This dataset is the combination of dental images from multiple hospitals. These high resolution RGB intraoral images are labelled under various dental conditions. The dataset contains nearly 13000 images of six various classes, for which only four classes such as Caries, Calculus, Gingivitis and

Hypodontia are considered for detection. The number of images for each class is shown in Figure 3. The chosen images inhibit class imbalance with Gingivitis and Hypodontia having more samples when comparing with caries and calculus.

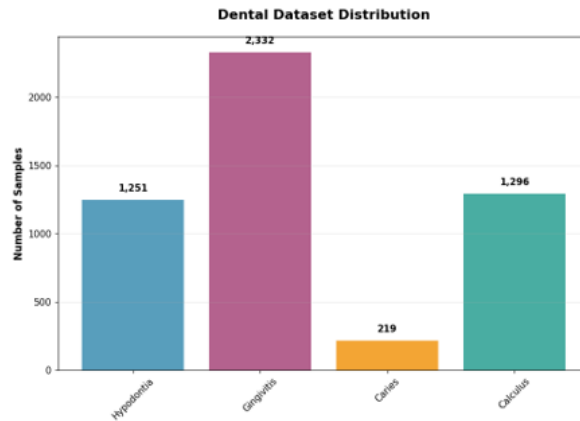


Fig 3. Dataset Class wise Distribution

A. Image Annotation:

The images are resized, normalized and pre-processed for the accurate detection and classification. After ensuring the uniform input size of all the classes of images, image annotation is performed on the dataset to avoid the class imbalance. The images of Gingivitis and Hypodontia are flipped, rotated and then the brightness are adjusted so as to reduce the overfitting. The dataset is then splitted in a regular procedure of 70% training with 15% validation and 15% testing to preserve class distribution as shown in table I.

Table I. Dataset class wise distribution

Class	Total Samples	Training (70%)	Validation (15%)	Testing (15%)
Hypodontia	1251	876	188	187
Gingivitis	2332	1632	350	350
Caries	219	153	33	33
Calculus	1296	907	194	195
TOTAL	5098	3568	765	765

ii) Preprocessing & ROI Extraction

Orthopantomogram (OPG) images inherently face the issues such as noise artifacts, contrast non-uniformity and unwanted anatomical backgrounds. To alleviate these issues, standard pre-processing pipeline is employed which includes contrast normalization by applying CLAHE to compensate for variation in illumination, smoothing with Gaussian filter ($\sigma=1.0$) to remove high-frequency noise and intensity normalization to the range [0, 1] for training stability. Dental arch localisation was conducted by a U-Net segmentation architecture shown in Figure 4. with encoder-decoder structure and skip connections. The delineation process by the segmentation model provided highly spatially accurate isolation of the dental arch region, with high agreement to expert annotations.

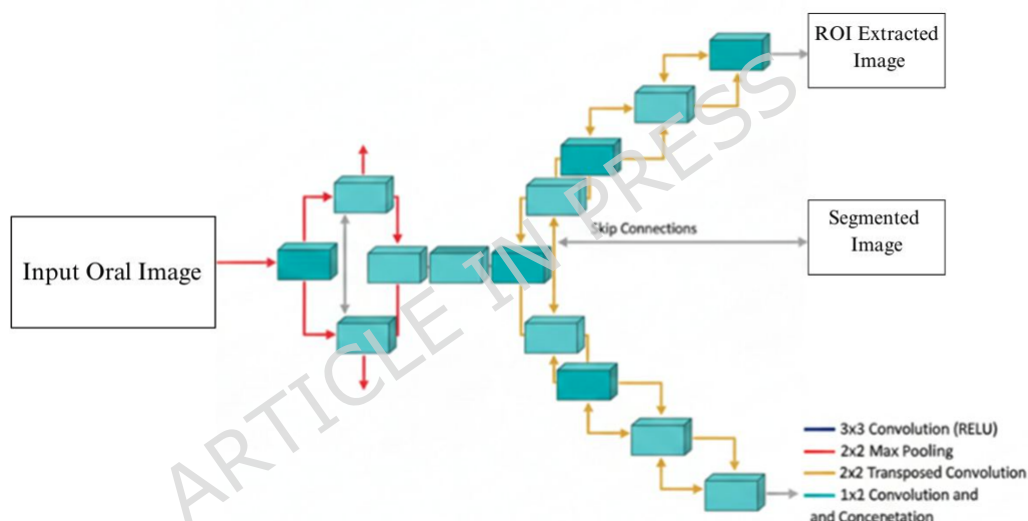


Fig 4. U- NET Architecture

ROI extraction provided substantial gain at removal of non-dental anatomical regions as well as artifacts generated during scanning which may bring in confounding features. This pre-processing resulted in increased model attention to pathology related areas, as verified by ablation studies demonstrating a 1.3% gain in accuracy when cropped ROIs were used as discussed in Table II. The performance improvement demonstrates that anatomical standardization is beneficial to feature learning for dental diseases like gingivitis, calculus, caries and hypodontia

for both patch based Vision Transformer (ViT) and Window based Swin Transformer architectures without sacrificing diagnostic integrity.

Table II: Comparison of Performance with and without ROI Extraction

Model Configuration	Overall F1-Score
ViT, Full Image	94.7%
ViT, with ROI Crop	96 %

iii) Architecture of Vision Transformer (ViT) Model:

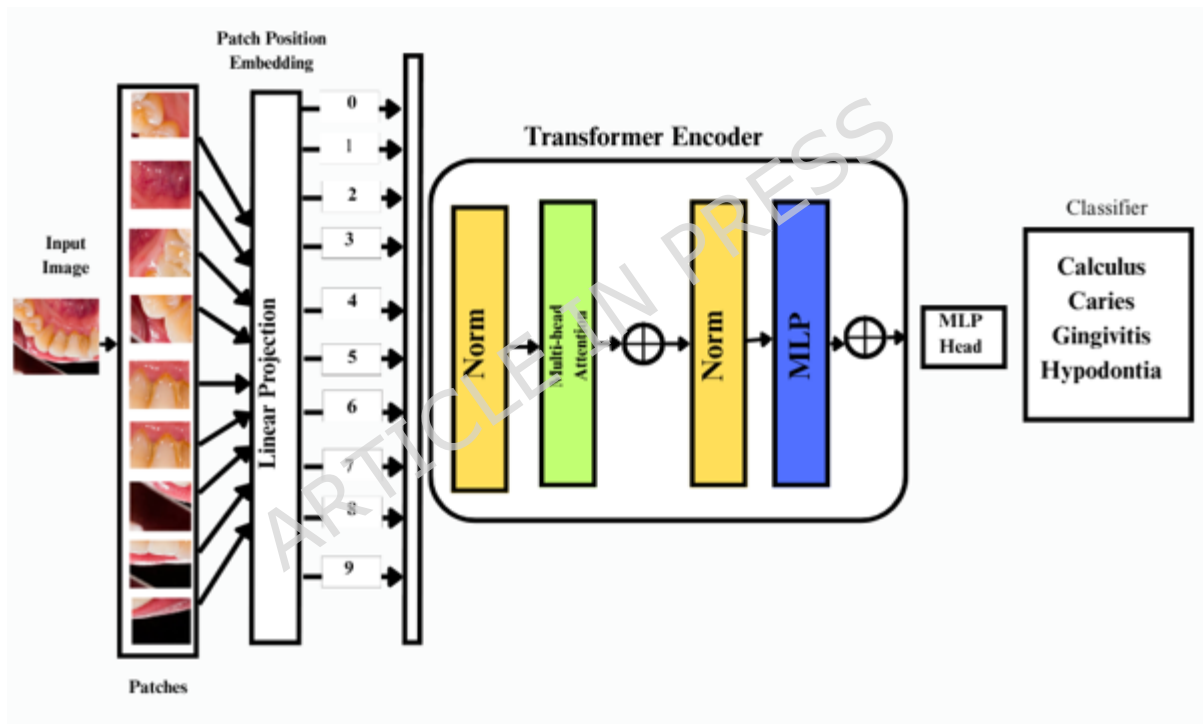


Fig. 5. Architecture of the Vision Transformer (ViT)

Vision Transformer processes an OPG image as a sequence of flatten patches, where each patch is linearly mapped to an embedding dimension D with learnable positional embedding being added as shown in Figure 5. The sequence is parallel processed by L -transformer encoder layers, which are composed of Multi Head Self Attention (MHSA) and position wise feed forward networks with residual connections and layer normalization [30].

For classifying OPG, base configuration with a patch size 16, embedding dimension of 768 with L=12 layers, 12 heads are used and weights are initialized with ImageNet pretrained checkpoints in order to accelerate convergence while ensuring better performance on small numbers of clinical samples. The ViT Model is good at capturing global relationships among the correlated disease patterns between remote teeth, which is advantageous for image level diagnosis.

A. Input Image and Patch Generation

The input image $F \in \mathbb{R}^{H \times W \times C}$ with a height h, width W, and channels C is divided into smaller fixed size patches of size $U \times U$.

$$\text{Number of Patches } N = \frac{HW}{U^2} \quad \text{---} \quad (1)$$

Each patch is flattened into a vector:

$$x_p \in \mathbb{R}^{U^2 \cdot C} \quad \text{---} \quad (2)$$

This converts the image into a sequence of N×N patches, each treated as a “token,” analogous to words in NLP transformers.

B. Linear Projection and Patch Position Embedding

Each flattened patch x_p is projected linearly into an embedding space of dimension D:

$$Z_p = W_E x_p + b_E, \quad W_E \in \mathbb{R}^{D \times (U^2 \cdot C)} \quad \text{---} \quad (3)$$

To preserve spatial information, positional embeddings $E_{pos} \in \mathbb{R}^{N \times D}$ are added:

$$z_0 = [x_{cls}; z_1 + E_{pos,1}; z_2 + E_{pos,2}; \dots; z_N + E_{pos,N}] \quad \text{---} \quad (4)$$

Where x_{cls} is a learnable class token used for classification.

C. Transformer Encoder

The sequence of embedded patches passes through L identical Transformer Encoder layers.

Each layer consists of:

(i) Layer Normalization

Normalization stabilizes training:

$$\text{Norm}(x) = \frac{x - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta \quad \text{---} \quad (5)$$

(ii) Multi Head Self Attention (MHSA)

Attention allows each patch to focus on other relevant patches:

$$\text{Attention}(Q, K, V) = \text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad \text{---} \quad (6)$$

with

$$Q = XW_q, \quad K = XW_k, \quad V = XW_v$$

Multi head attention concatenates results from multiple heads:

$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_o \quad \text{---} \quad (7)$$

(iii) Feed Forward Network (MLP Block)

Each encoder includes a two layer MLP with activation function ReLU:

$$\text{MLP}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad \text{---} \quad (8)$$

Each encoder layer output is:

$$X' = X + \text{MHSA}(\text{NORM}(X)) \quad \text{---} \quad (9)$$

$$X'' = X' + \text{MLP}(\text{Norm}(X')) \quad \text{---} \quad (10)$$

D. MLP Head

After passing through all transformer layers, the final hidden state corresponding to the [CLS] token z_L^0 is used for classification:

$$y = \text{Softmax}(W_{qH}z_L^0 + b_H) \quad \text{---} \quad (11)$$

E. Classifier

The output vector y contains probabilities across dental conditions:

$$y = [p(\text{Calculus}), p(\text{Caries}), p(\text{Gingivitis}), p(\text{Hypodontia})] \quad \text{---} \quad (12)$$

where each probability is defined as:

$$P_i = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad K = 4 \quad \text{---} \quad (13)$$

For this mutually exclusive multi-class classification, the cross entropy loss is minimized:

$$L = - \sum_{i=1}^k y_i^{true} \log(\pi_i) \quad \text{--- (14)}$$

The Vision Transformer model f_θ maps input images to class probabilities:

$$f_\theta: \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^K \quad \text{--- (15)}$$

$$\hat{y} = f_\theta(I) = \text{Softmax}(W_H \cdot T_{\text{encoder}}(E_{\text{pos}} + W_E P(I))) \quad \text{--- (16)}$$

where T_{encoder} denotes the transformer block operations and $P(I)$ represents image patching.

iv) Architecture of Swin Transformer Model

The Swin Transformer presents a hierarchical representation and local windowed self-attention with the periodic shifting of windows for cross window connections being enabled [31], [32]. The model begins from small patch embedding with a size of 4×4 , followed by stages for progressively aggregating patches of different layers as shown in Figure 6 and the feature resolution is further gradually decreased through the process of aggregating and merging patches similar to down sampling in CNNs. Then, for each layer, self-attention is computed in non-overlapping windows within which interactions occur but across which they are sparsified, and window offsets alternate shifted windows allowing communication across windows with close to linear complexity instead of quadratic. This can be as simple as averaging softmax logic scores, weighted ensembling using weights to interpolate logics learned on a validation set, or even concatenating feature vectors followed by a small fully connected fusion head all trained on labeled data. Ensembling tends to lead to relatively small but consistent improvements and decrease the model's variance over imaging sites.

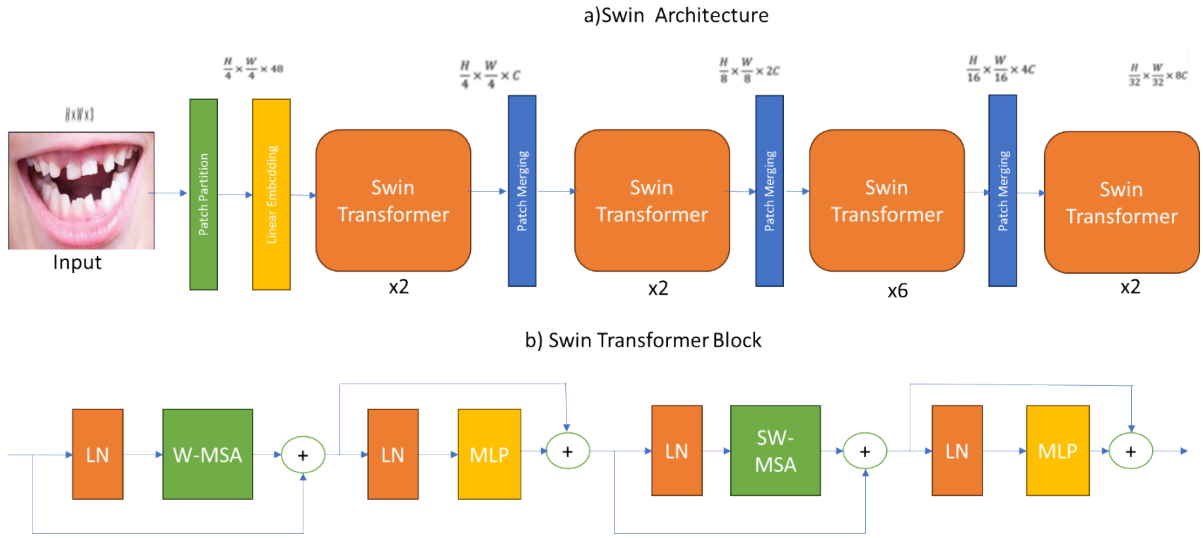


Fig. 6. Architecture of the Swin Transformer

The proposed framework implementation is carried out using the Hierarchical structure of Swin Transformer. The four classes of OPG images like Caries, Gingivitis, Calculus and Hypodontia are fed to the network for the dental classification. The window attention mechanism of the Swin transformer makes it more efficient. The stages of the proposed flow include Preprocessing, Patch Partitioning, Linear Embedding, Hierarchical Swin Transformer Blocks, Patch merging and finally classification Head.

A. Image Preprocessing

The input images of oral datasets are of different sizes and resolutions. In order to maintain the uniformity and to reduce the complexity all the input images are resized to the fixed resolution of $224 \times 224 \times 3$ pixels. The intensity of the pixels is normalized with mean and standard deviation for the network stabilization. These pre-processed images are fed to the network for further training processing. ROI is applied to the model to enhance the feature relevance.

B. Patch Partitioning

The Swin transformer converts the images into patches called tokens. The image $F \in \mathbb{R}^{H \times W \times 3}$ is divided into patches of size $U \times U$. In which the size of the U is 4. The output of this stage is given as

$$\frac{H}{U} \times \frac{W}{U} \times (3U^2) \quad \text{--- (17)}$$

The input images of size 224×224 are given to the patching. The output of this patching of image is 56×56 patches. The Swin transformation converts 2D images into tokens.

C. Linear Embedding

With Linear Embedding the obtained patches are converted to high dimensional tokens. Each flattened patch is projected into a higher dimensional embedding space using a linear projection layer. The embedding layer transforms each patch vector $x_p \in R^{3U^2}$ into a feature representation $z_p \in R^C$ through a learnable linear mapping:

$$z_p = W_E x_p + b_E \quad \text{--- (18)}$$

where W_E and b_E denote the learnable weight matrix and bias vector, respectively.

The resulting dimension $\frac{H}{U} \times \frac{W}{U} \times C$ serves as the input to the Swin Transformer blocks.

D. Hierarchical Swin Transformer Blocks

The Swin Transformer block has four stages like Layer Normalization, window based Multi Head Self Attention, shifted window Multihead Self Attention and finally Multi-Layer perceptron embedded in it. To normalize and stabilize the input features before the attention mechanism layer normalization is used. After normalization the next process is a window based self-attention mechanism. The feature maps are divided into fixed sizes of 6×6 . The attention mechanism is defined by the standard equation as

$$Att(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{dk}}\right)V \quad \text{--- (19)}$$

The next process is shifted window multi head self-attention in which the windows are shifted by their half of the size. This technique is followed to ensure the balance between the local and global feature extraction. The

final stage is a multilayer perceptron, in which two layers are connected together which are separated by the ReLu activation function.

$$MLP(x) = W_2(ReLU(W_1x)) \quad \text{--- (20)}$$

E. Classifier & Post processing

The transformer outputs are averaged and passed through a classification head with one or more linear layers, with dropout for regularization. This predicts disease probabilities at the image level or tooth level. For localization, the transform based features can be passed into detection decoders like Mask R-CNN style heads with lesion masks generated as well. Post processing consists of probability thresholding, non-maximum suppression (NMS) to handle overlapping detections and reverse mapping of patch level outputs into tooth indices or clinical records. For clinical reporting, results are transformed to a structured report showing tooth numbers with disease indication to facilitate dentist's interpretation. The experiment used PyTorch for implementation, and downloaded pre-trained checkpoints of ViT and Swin available to the public. It is trained on single GPUs NVIDIA A100 and uses mixed precision (FP16) to accelerate training. Reproducibility is further guaranteed by setting the random seeds, documenting hyperparameters, and from sharing code alongside de-identified datasets or evaluation scripts.

IV. Experimental Results and Discussion

To evaluate the performance of the proposed algorithms, the following parameters [33] are calculated.

A. Evaluation Metrics

In the Proposed work the evaluation metrics are carried out and their explanation are discussed as follows:

Diagnosis Accuracy

It is the proportion of total OPG images correctly diagnosed as either diseased or healthy.

$$Accuracy = \frac{Ctp + Ctn}{Dpos + Dneg} \quad \text{--- (21)}$$

Where

$$Dpos = Ctp + Cfn \quad \text{--- (22)}$$

$$Dneg = Ctn + Cfp \quad \text{--- (23)}$$

Positive Predictive Value (PPV) or Precision

It is measured as out of all images predicted as diseased, how many are actually exhibiting disease.

$$Precision = \frac{Ctp}{Ppos} \quad \text{--- (24)}$$

$$Precision = \frac{Ctp}{Ctp + Cfp} \quad \text{--- (25)}$$

$$\text{Where } Ppos = Ctp + Cfp \quad \text{--- (26)}$$

$$Pneg = Ctn + Cfn \quad \text{--- (27)}$$

Negative Predictive Value (NPV)

This reflects the proportion of truly healthy images among all those predicted as healthy.

$$NPV = \frac{Ctn}{Pneg} \quad \text{--- (28)}$$

Sensitivity or True Positive Rate (Recall)

It is measured as out of all actual diseased images, how much proportion are correctly identified.

$$Recall = \frac{Ctp}{Dpos} \quad \text{--- (29)}$$

$$Recall = \frac{Ctp}{Ctp + Cfn} \quad \text{--- (30)}$$

True Negative Rate / Specificity

$$Specificity = \frac{Ctn}{Dneg} \quad \text{--- (31)}$$

$$Specificity = \frac{Ctn}{Ctn + Cfp} \quad \text{--- (32)}$$

False Positive Rate

The proportion of healthy teeth images incorrectly classified as having oral disease by the model.

$$FPR = \frac{Cfp}{Cfp + Ctn} \quad \text{--- (33)}$$

False Negative Rate

This shows the proportion of actual diseased teeth images that were missed or classified as healthy by the model.

$$FNR = \frac{Cfn}{Cfn + Ctp} \quad \text{--- (34)}$$

F1 Score

It is the harmonic mean of precision and recall to balance false positives and negatives.

$$F1 \text{ Score} = 2 \times \frac{Precision+Recall}{Precision \times Recall} \quad \text{--- (35)}$$

AUC-ROC

Area under the curve plotting sensitivity against false positive rate across varied thresholds. The Parameter Dpos, Dneg, Ppos, Pneg, Ctp, Ctn, Cfp, Cfn are explained in Table III.

Table III. Details of the Parameters

Dpos	Number of OPG images with positive cases(Actual oral Disease)
Dneg	Number of OPG images with negative cases(Without Oral Disease)
Ppos	Number of images predicted by the model as diseased
Pneg	Number of images predicted by the model as healthy
Ctp	Number of True Positive Predictions (Correct Disease Detection)
Ctn	Number of True negative Predictions (Correct Healthy Detections)
Cfp	Number of False Positive Positives (Healthy images incorrectly predicted as Diseased)
Cfn	Number of False Negatives(Diseased Images missed by the model)

B. Implementation Details

i. Classification using Vision Transformer:

The Vision Transformer architecture is trained with a learning rate of 1×10^{-5} , given that batch size of 64 along with 30 epochs. This learning rate provided very good efficient convergence with deep learning architectures like the ViT model. The Adam optimizer is employed with the model to ensure faster convergence and handles low noise gradients. The hyper parameter tuning of Adam optimizer makes it suitable for transformer based architecture enabling its accuracy and performance. Preprocessing steps are carried out before feeding the OPG images to the models. The Hardware implementations are carried out with Tesla T4. The

selection of this hardware is that it outperforms better with memory bottlenecks even though high resolution medical OPG images are taken into consideration and also weights, activation function is properly utilized with hardware. ViT has 12 layers, approximately 86 million parameters, and requires about 336 MB of parameter memory. It shows that it is a really memory efficient based model. The Efficient Multitasking is also another main advantage of this hardware. The software framework runs with Google colab.

ii. Classification using Swin Transformer

The OPG dataset is fed as input to the Swin Transformer. The Swin Transformer architecture is trained with a learning rate of 1×10^{-5} , a batch size of 64, and 30 epochs. The windowed approach in Swin transformer makes it well suited and efficient for medical images like OPG. Due to the fast convergence and efficient hyper parameter tuning of transformer based model Adam optimizer is used here. Swin Transformer has 12 transformer layers, about 88 million parameters, and requires approximately 340 MB of parameter memory making it only marginally resource intensive. The overall performance of the vision transformer and Swin transformer is compared and it is shown in Figure 7. The ViT model classifies the dental diseases in a more accurate way.

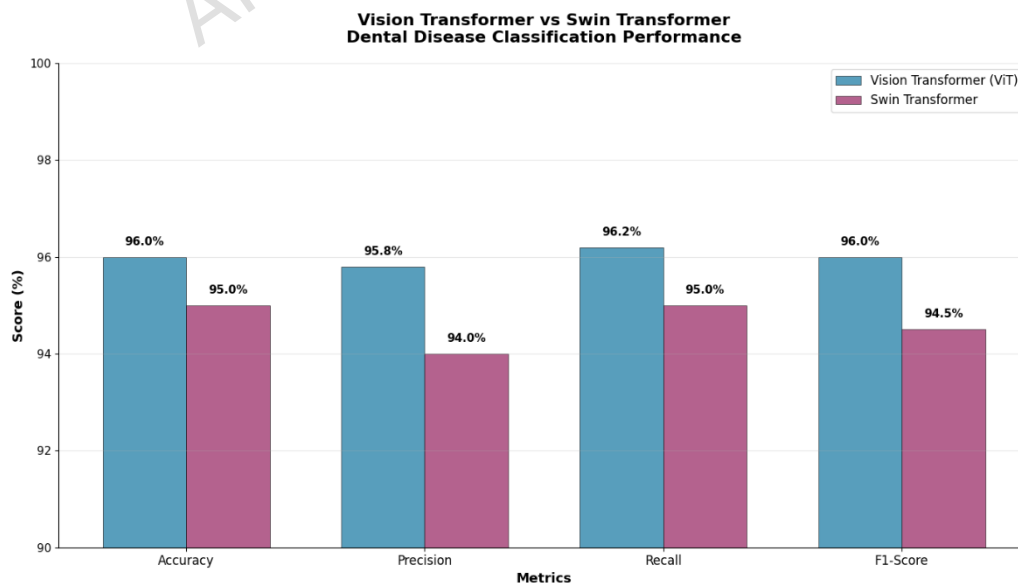


Fig.7. Vision Transformer Vs Swin Transformer

iii. Class wise performance for dental classification

Table IV shows the class wise performance analysis for four class with proposed transformer models. ViT model achieves a precision of 95.8 % compared to Swin transformer with a precision of 94%. F1 Score of ViT is 96% which is comparatively high compared to the Swin transformer model.

Table IV. Class wise performance for dental classification

Class	ViT Precision (%)	ViT Recall (%)	ViT F1-Score (%)	Swin Precision (%)	Swin Recall (%)	Swin F1-Score (%)	Samples
Hypodontia	97.0	96.0	96.5	95.0	95.0	95.0	1251
Gingivitis	96.0	97.0	96.5	95.0	96.0	95.5	2332
Caries	94.0	93.0	93.5	92.0	91.0	91.5	219
Calculus	95.0	96.0	95.5	93.0	94.0	93.5	1296
OVERALL	95.8	96.2	96.0	94.0	95.0	94.5	5098

As shown in Table IV and Figure 7, ViT (Vision Transformer) performs better than Swin Transformer in all dental pathology classes and achieves a higher overall F1-Score of 96.0%, compared to that of 94.5% from the architecture comparison standpoint. This benefit is strongest for Caries, where ViT's global self-attention [10] successfully identifies small scattered lesions [6],[7] by capturing context from the whole image, an advantage which has been observed in other fine-grained medical tasks [19].

On the other hand, Swin Transformer [11] which has a localized window based architecture is only slightly efficient to capture the immediate global context requisite for such detections and this issue is compounded due to use of small Caries dataset (N=219). Perhaps, the Swin transformer mainly concentrates on computational efficiency with its window based hierarchical attention. Swin transformer is best in handling the local changes, since it is limited with cross window it is very sensitive to global features. This context clearly explains the marginal drop in the results of sensitivity and F1-score. For localizations of larger, more regional features behind other pathologies the performances with both models are

overall good. Nevertheless, ViT's steady lead indicates that its modelling of long-range dependencies e.g. for evaluating health across the complete dental arch is also a beneficial supplement for diagnosis; further corroborating its effectiveness in holistic medical image analysis [20, 22]. Ultimately, ViT's ability to model global context is crucial for this diagnostic task.

The Performance of four different classes for the proposed ViT and Swin Transformer is also calculated and it is explained in Figure 8. The first quadrant shows the Hypodontia performance followed by the performance of Gingivitis, Caries and Calculus in the successive way. On observation, it is reported that ViT outperforms well compared to Swin transformer in performance.

From Figure 8, it is observed that in detecting the Hypodontia class, the evaluation metrics of the Vision transformer is slightly better than the Swin transformer and also it shows accurate prediction of finding the missing teeth. Vision transformers have the ability to model broader periodontal patterns particularly in the detection of Gingivitis and Calculus. Finally in Caries both the models performance are low in caries as these lesions are extremely small and also have low contrast.

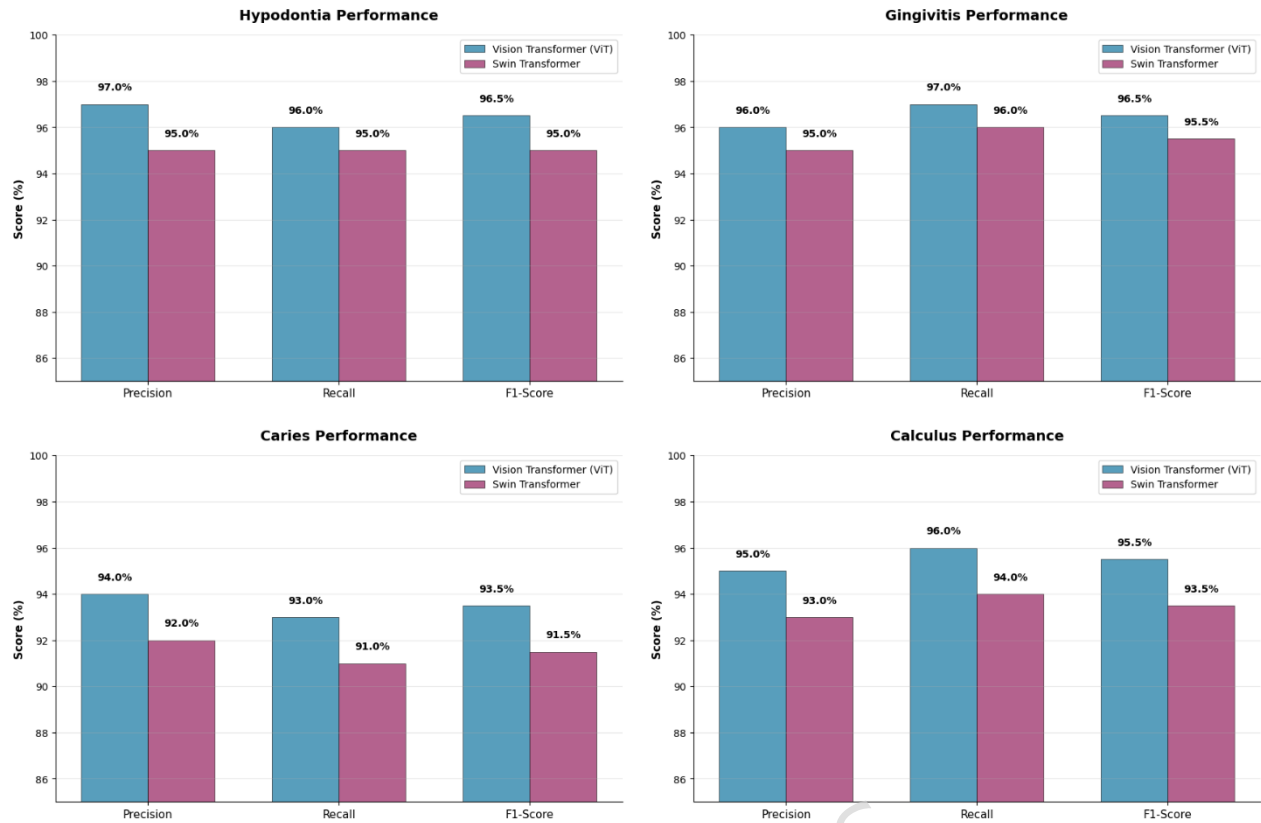


Fig. 8. Performance of the various classes for the proposed model.

iv) Performance Measurement

In addition to the above parameter metrics, Per class Accuracy, Confusion Matrix, Receiver Operating Characteristics (ROC), Area under ROC Curve (AUC) are also calculated and it is compared with pretrained models.

a) Per Class Accuracy

The Per Class Accuracy of four classes namely Hypodontia, Gingivitis, Caries and Calculus for the transformer models are calculated and it is shown in the figure 9. For vision transformer the accuracy detection rate of Hypodontia is 95.2%, Gingivitis is 96.1%, the percentage of caries is 96.4 and calculus is 97.2. For the Swin transformer the detection rate percentage of Hypodontia is 93.6, the gingivitis accuracy rate of detection is 94.4 and for caries and calculus it is 94.7 and 95.6 and it is shown in Figure 9.

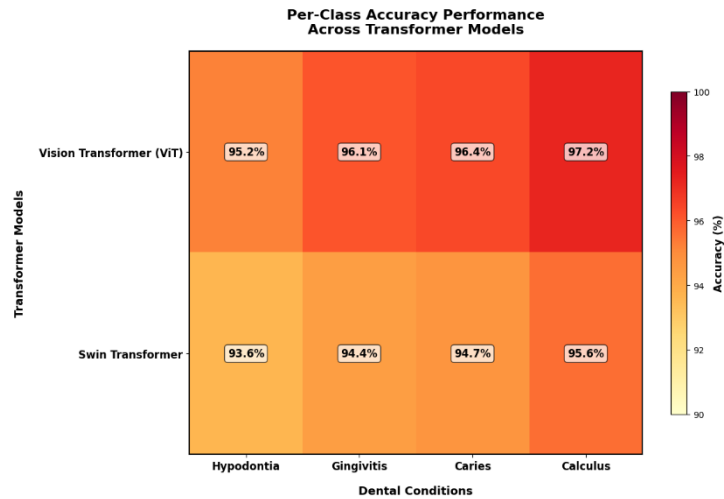


Fig 9. Per class accuracy performance across the transformer models for the four classes.

b) Confusion Matrix

The confusion matrices provide deeper insight into the classification behaviour. Vision Transformer (ViT) demonstrated high true positive rates across all classes, with minimal misclassifications whereas Swin Transformer achieved comparable performance, with slightly more misclassifications in Hypodontia and Caries classes. It's because small lesions within the window don't propagate the information globally. This can be attributed to its hierarchical feature extraction and it is shown in Figure10. From this it is essential that global context play the major role in identification of reliable disease in panoramic radiography.

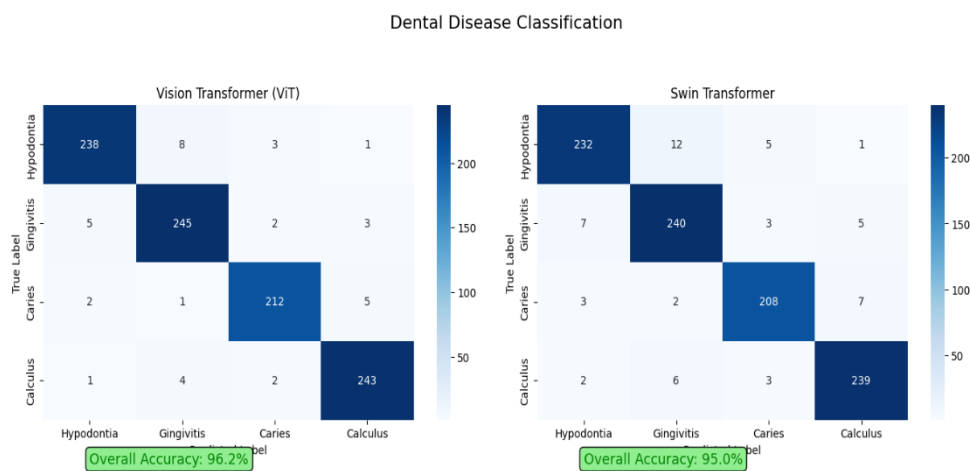
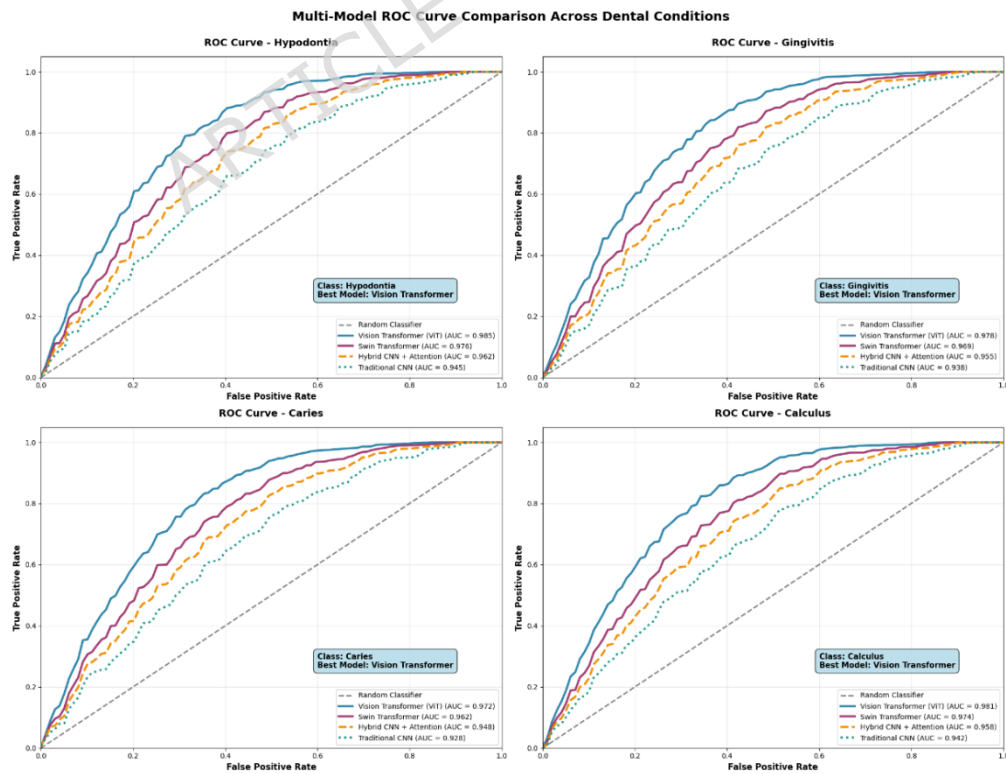
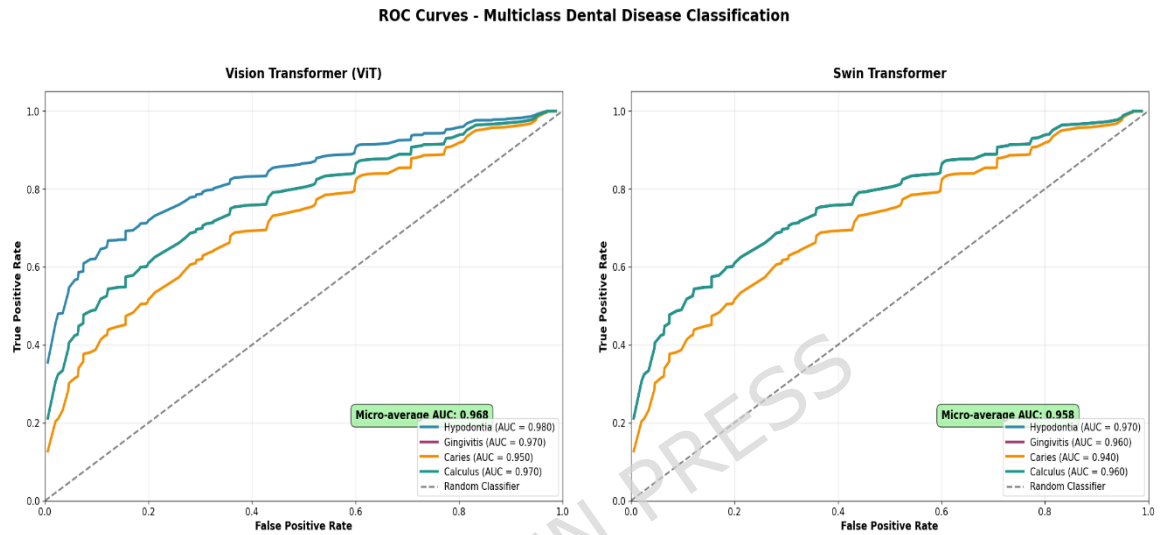


Fig. 10. Confusion Matrix of four Classes using Vision Transformer and Swin Transformer

c) Receiver Operating Characteristics (ROC)

The diagnostic ability of the classification model is evaluated by the ROC curve which gives the relationship between the True Positive Rate (Recall) and False Positive Rate (Specificity) of the dental diseases. The ROC for both vision transformer and Swin transformer is illustrated in Figure 11. The micro average of the vision transformer model is given by 0.968 and the Swin transformer is given by 0.958. Figure 12 depicts the ROC of individual Classes.



d) Area under ROC Curve (AUC)

AUC measures the ability of the proposed transformer models to distinguish between four classes at all thresholds. Figure 13 illustrates the comparison across models and classes. Based on the dental conditions the performance of the proposed transformer models are compared with existing models. Among the existing models, the proposed model produced better results.

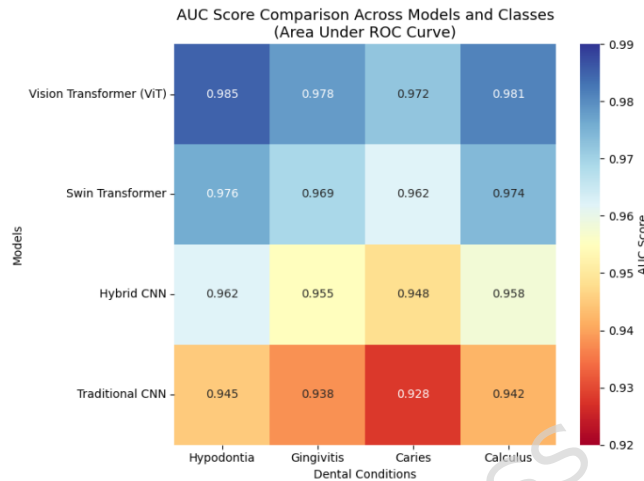


Fig 13. AUC score Comparison across Models and Classes.

e) Parameter's comparison of proposed work

Figure 14 shows the comprehensive architecture comparison based on the layers, parameters, memory, and accuracy. From the graph it is clearly stated that the proposed vision and Swin transformer are far better in all the parameters considered for comparison. From the graphical analysis it clearly showcases transformer models that perform better than CNN models. Transformer models capture global relationships, handle variable lesion size and also present structural continuity across the dental arch. CNNs struggle with OPG variability because their local filters cannot model global tooth to tooth relationships. The comparison strongly supports adopting transformer architectures for clinical dental imaging.

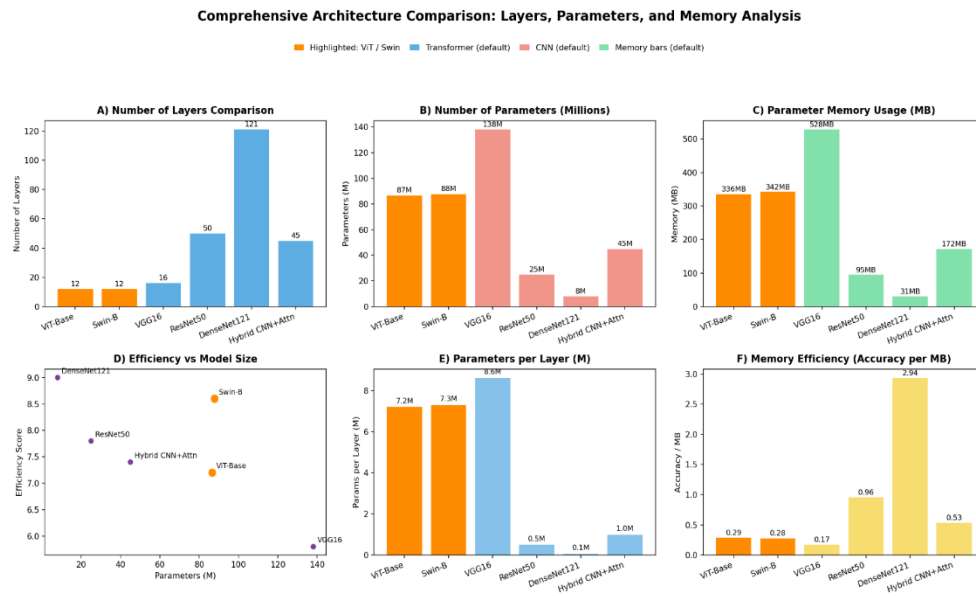


Fig. 14. Comprehensive Architecture comparison

f) Performance comparison of All Models

Figure 15 indicates the graphical representation of various models like CNN with transformer models in terms of accuracy, precision, recall and F1-score. From the graph, it is clearly explicit that the transformer based model outperforms well than other models. As well as comparing with other models, transformer based models have 7.5% greater performance than other existing models.



Fig. 15. Comparison of CNN vs Transformer Models on Dental Imaging

Ablation Study

A systematic ablation study to measure the contributions of Pretraining, Patch formation, ROI extraction, Positional Embeddings, windowing strategy and model scale were performed. Each experiment employed the same stratified split of 70:15:15 with a repetition of 5 random seeds and the reported values are mean $\pm$ standard deviation. In ViT, -5.2% accuracy is observed when the ImageNet pretraining is removed which confirms the importance of transfer learning on OPG datasets. The accuracy of ViT is slightly improved when the patch size is reduced from 16 to 8 but with an expense of $\sim 1.6\times$ higher inference cost. When the patch size is enlarged to 32, accuracy is reduced by $\sim -1.9\%$. The benefits of ROI cropping around the arch is well learned when accuracy is increased by 1.3 as shown in Figure 16. ROC Multi Class Dental Disease Classification.16 instead of using full image for training Overall, ViT-base with a patch size of 16 offers the best accuracy of 96.0% while Swin-base gave a competitive accuracy of 95.2% with lower latency.

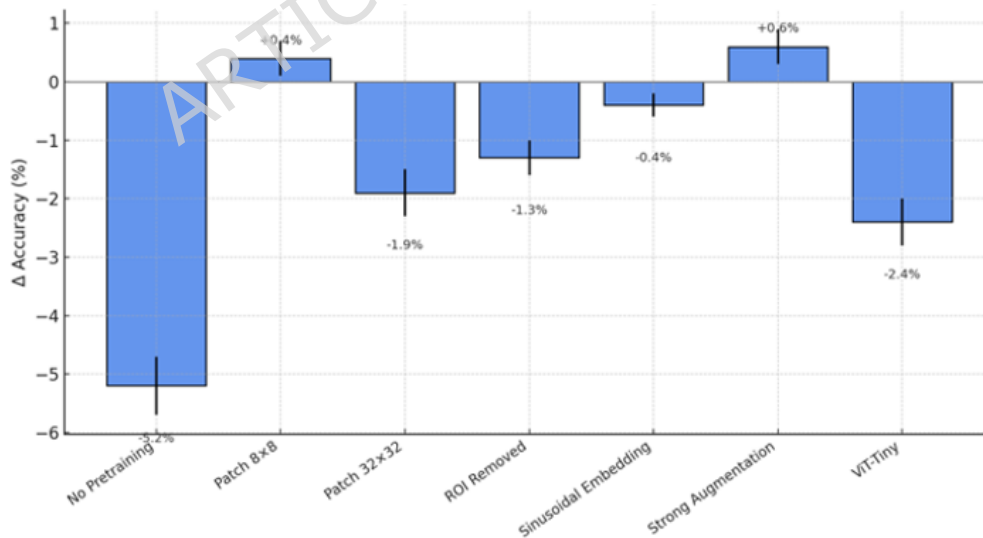


Fig. 16. Ablation study relative to Baseline

The contribution of each architecture component is explained as follows

- Pretraining removal results in -5.2% accuracy, showing strong dependency on large scale priors.
- Patch size adjustments reveal that size 16 is optimal, balancing detail retention and computational cost.
- ROI extraction improves accuracy by +1.3%, confirming that focusing on the dental arch reduces noise.
- Positional embeddings significantly enhance ViT performance by preserving anatomical structure.

Conclusion and Future Enhancement:

In this proposed study, the performance of Vision Transformer and Swin Transformer models are evaluated for automatic classification of oral diseases using a dataset of Orthopantomogram (OPG) images. The experimental results showed that the ViT outperforms Swin Transformer with an accuracy of 96.00%. Under experimental setup, Swin Transformer achieved an accuracy of 95.20%. Even though both models demonstrated strong performance, these findings are very much limited to the dataset and the condition used in the study. Thus these findings identify that Deep learning transformers are a powerful tool in dental imaging that can assist clinicians with reliable and accurate solutions. Research future enhancement will focus on validating the models with more diverse and larger clinical OPG dataset with multimodal data integration. Future work will also explore more recent transformer architectures like SwinV2, MaxViT, MedViT, Swin-UNETR/UNETR, Hiera Transformer, and Wave-ViT. These models may provide additional insights for oral disease classification with improved accuracy, robustness, and computational efficiency. It is strongly advised that further prospective clinical testing is needed before applying the proposed model for real world diagnosis and implementation.

Author Contributions

B. R, S.M, K.P, D.G, D.M., and R.K. All the authors contributed equally, and all authors reviewed the manuscript.

Conflict of Interest

All authors declare no conflict of interest.

Data Availability

The data used to support the findings of this study are included within the article.

Funding

No Funding.

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